

Exercise 2.1: Safety Vocabulary

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:

1. Loss
2. Hazard
3. Risk

1. Loss

A loss is any adverse consequence — something bad that actually happens and that we want to prevent. Think of it as the worst-case outcome you're ultimately trying to avoid. loss of human life, injury, material damage, or even a violation of law resulting in fines.

Autonomous driving example: The self-driving car fails to brake in time and collides with a cyclist, resulting in the cyclist's injury. The injury itself is the loss — the actual, realised harm.

2. Hazard

A hazard is a system state or condition that can lead to a loss under certain circumstances. The important nuance here is that a hazard doesn't guarantee a bad outcome — it just makes one possible.

Autonomous driving example: The vehicle's perception system fails to detect a pedestrian stepping onto the road. The vehicle is now in a dangerous state — moving toward an obstacle it doesn't know is there. This is the hazard. It might still stop in time (no loss), or it might not. Either way, the dangerous condition is the hazard.

3. Risk

Risk is how seriously we should worry about a given hazard.

$$\text{Risk} \approx P(\text{hazard}) \times P(\text{loss} \mid \text{hazard}) \times \text{Severity}$$

How likely is the hazardous situation to occur? If it does occur, how likely is it to cause a loss? And if a loss does occur, how bad would it be? Risk analysis means answering those three questions together.

Autonomous driving example: Consider the hazard "vehicle commands acceleration when an obstacle is within stopping distance." The risk would be assessed as: How often does this happen? (probability of the hazard) → If it happens, does a collision actually occur? (probability of loss given hazard) → If a collision occurs, is it at low speed in a parking lot or at high speed on a motorway? (severity). A high-speed scenario with a vulnerable road user would carry very high risk; a slow-speed bump in a car park would be much lower.

the loss is what you're trying to prevent, the hazard is the dangerous condition that could lead there, and the risk is a quantified measure of how much that hazard should concern you – combining both probability and severity.

STPA: System-Theoretic Process Analysis

STPA Step 1: Define Purpose and Identify Losses, Hazards, and ODD

Exercise 2.2: ODD Specification

Based only on the system description in the course Roadmap (you have not yet seen the training data), define the Operational Design Domain (ODD) for the CARLA system.

Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed.

For each dimension, state:

1. The operating conditions (where the system is considered valid).
2. The non-operating conditions (where it is not).
3. How an ODD violation could be detected at runtime.

Hint: You are defining the ODD before looking at the data. In Exercise Sheet 3, you will compare this ODD to the actual training data and (possibly) discover gaps.

Dimension 1 – Weather

Operating conditions (valid): Clear, dry weather, sunny skies with no precipitation, no fog, and no significant atmospheric obscurance. Light overcast might be tolerable

Non-operating conditions (invalid): Rain, fog, snow, heavy overcast that changes scene brightness, or any precipitation that affects the camera lens or image quality. The roadmap explicitly calls out rain onset and sudden cloud cover as conditions that can occur mid-drive and that lie outside the intended operating envelope.

Runtime detection: Monitor the raw image for sudden brightness drops, rain on the lens, or reduced contrast/sharpness across frames. A statistical change detector (e.g. mean luminance shift over a rolling window) or a dedicated weather classifier could flag this. Since there is no

wiper sensor or weather station on the vehicle, all detection has to come from the camera signal itself.

Dimension 2 – Lighting

Operating conditions (valid): Daytime with adequate ambient light – natural daylight without direct sun glare, no artificial-only lighting environments. The models were trained on sunny daytime data, so well-lit, naturally illuminated scenes are the valid domain.

Non-operating conditions (invalid): Night-time driving, dusk/dawn with low sun angles, tunnels, underground car parks, or any environment where primary illumination is artificial. Strong direct glare from a low sun into the camera lens also constitutes a lighting ODD violation even if it technically occurs during daytime.

Runtime detection: Compute average image luminance per frame. A threshold below a calibrated minimum indicates insufficient lighting. Similarly, a very high luminance region concentrated in the image centre could indicate direct sun glare. Both can be flagged as ODD violations. Clock-based triggers could serve as a secondary soft signal if a real-time clock is available.

Dimension 3 – Camera Condition

Operating conditions (valid): Clean, unobstructed lens. Camera rigidly mounted behind the windshield with no dirt, water droplets, cracks, or smearing on the lens or windshield in the camera's field of view. Frame rate stable at 10 Hz.

Non-operating conditions (invalid): Lens or windshield obscured by mud, water splashes, frost, condensation, bird droppings, or stickers. Camera physically displaced . Frame drops or latency that reduce effective frame rate below the expected 10 Hz.

Runtime detection: Blur metrics on consecutive frames (e.g. Laplacian variance) can detect sudden loss of sharpness across the whole image, which is a strong signal of lens contamination. Temporal consistency checks – if the image appears nearly identical across many frames despite vehicle movement – can flag camera freezes. Frame-rate monitoring at the system level can catch dropout events.

Dimension 4 – Scene Type

Operating conditions (valid): Public urban roads with mapped intersections and cars, pedestrians, traffic lights. Structured environments with lane markings, kerbs, and standard road furniture. The roadmap says the vehicle is tested on "public urban roads" with "mapped intersections".

Non-operating conditions (invalid): Motorways or high-speed roads, off-road terrain, construction zones with non-standard road layouts, underground environments, multi-level car parks, unmapped intersections, or any scenario involving actors not represented in the training data (e.g. trams, horses, unusual obstacles). The system has no map-matching capability described, so in practice it cannot verify whether a junction is "mapped" without additional tooling – this is an important runtime gap.

Runtime detection: GPS/map-based geofencing is the most robust approach – if the vehicle leaves a pre-approved map region, an ODD violation is raised. Without that, anomaly detection on the scene content (e.g. detecting unusual image statistics inconsistent with the training distribution) can provide a camera-only proxy. Sudden appearance of very large or very small objects outside the expected size distribution of cars and pedestrians could also be a soft signal.

Dimension 5 – Vehicle Speed

Operating conditions (valid): 0 to 50 km/h, as explicitly stated in the roadmap: "speeds up to 50 km/h". This also aligns with typical urban speed limits and the physics of emergency braking with a single camera and no redundant sensors.

Non-operating conditions (invalid): Any speed above 50 km/h. At higher speeds, the stopping distance grows non-linearly while the reaction time window (one frame every 100 ms at 10 Hz) stays constant – so the system's design assumptions about braking distances break down. Reverse driving is also outside the ODD since the camera is forward-facing only.

Runtime detection: Vehicle speed is directly available from the drive-by-wire actuator system (wheel encoders or odometry). A simple threshold check – speed > 50 km/h – immediately flags a violation. This is one of the easiest dimensions to monitor reliably since it doesn't require any camera-based inference.

Exercise 2.3: Losses

STPA begins by identifying the losses that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.

For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

L1	Injury or death of a pedestrian caused by the AV failing to detect or react to a person in its path	Human life is the highest-priority value in any safety case; a missed pedestrian detection leading to a collision is the central risk this system exists to prevent
L2	Collision with another vehicle, causing injury to occupants or third parties and material damage	The AV shares public urban roads with other road users. A failure to detect a vehicle ahead and brake appropriately directly endangers both the AV's human operator and others — with no sensor redundancy, a single classifier failure is enough
L3	Violation of traffic rules -e.g. running a red light, leading to intersection collision or legal liability	Failure to detect a traffic light or misclassify its state can cause the vehicle to enter an intersection unlawfully, creating dangerous cross-traffic conflicts and exposing the operator and developers to legal and regulatory consequences
L4	Loss of control due to ODD violation (e.g. rain, night-time, or speed > 50 km/h) where the system operates beyond its valid domain, resulting in harm	The models were trained exclusively on sunny daytime data with no sensor redundancy; if the system continues to make autonomous decisions outside its ODD, all perception outputs are unreliable and any resulting harm is entirely preventable — the human operator is the only fallback and may not intervene in time

Exercise 2.4: Hazards

A hazard is a system state that, combined with worst-case environmental conditions, can

lead to a loss. Hazards are properties of the system, not of individual components.

1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

Fill in the following table:

H 1	The AV is moving while a pedestrian is within its stopping distance and the pedestrian detector outputs "not present" (false negative)	L-1, L-4	Medium — the model was trained only on sunny daytime data; any ODD deviation (rain, glare, occlusion) increases	High — direct collision with a vulnerable road user; likely fatal or seriously injuring
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			miss-rate significantly	
H 2	The AV is approaching a vehicle ahead at unsafe closing speed while the vehicle detector outputs "not present" (false negative)	L-2, L-4	Medium – similar distribution-shift vulnerability as H-1; dense urban traffic increases exposure frequency	High – rear-end or frontal collision; risk of injury to operator and third parties, major material damage
H 3	The AV crosses or enters an intersection while the traffic-light detector outputs "no red / stop not required" when the light is actually red	L-3, L-2, L-1	Medium – classifier trained on limited simulator data; unusual lighting angles, partial occlusion, or lens glare can confuse the detector	High – cross-traffic conflict at speed; potential T-bone collision injuring multiple parties
H 4	The AV is operating outside its ODD (e.g. night-time, rain, speed > 50 km/h) while the system continues to issue autonomous control commands as if conditions were normal	L-1, L-2, L-3, L-4	High – the roadmap explicitly states weather and lighting can change mid-drive (rain onset, cloud cover, low sun angle); there is currently no ODD monitor deployed	High – all three perception models become unreliable simultaneously with no sensor redundancy and only a potentially slow-reacting human operator as fallback
H 5	The AV commands acceleration or fails to decelerate when an obstacle is within stopping distance due to a planning module misinterpretation of perception outputs	L-1, L-2	Low – the planner logic is rule-based and well-defined, but edge cases (e.g. conflicting outputs from two classifiers) could produce unexpected behaviour	High – direct collision at potentially significant speed with no last-resort braking layer other than the human operator

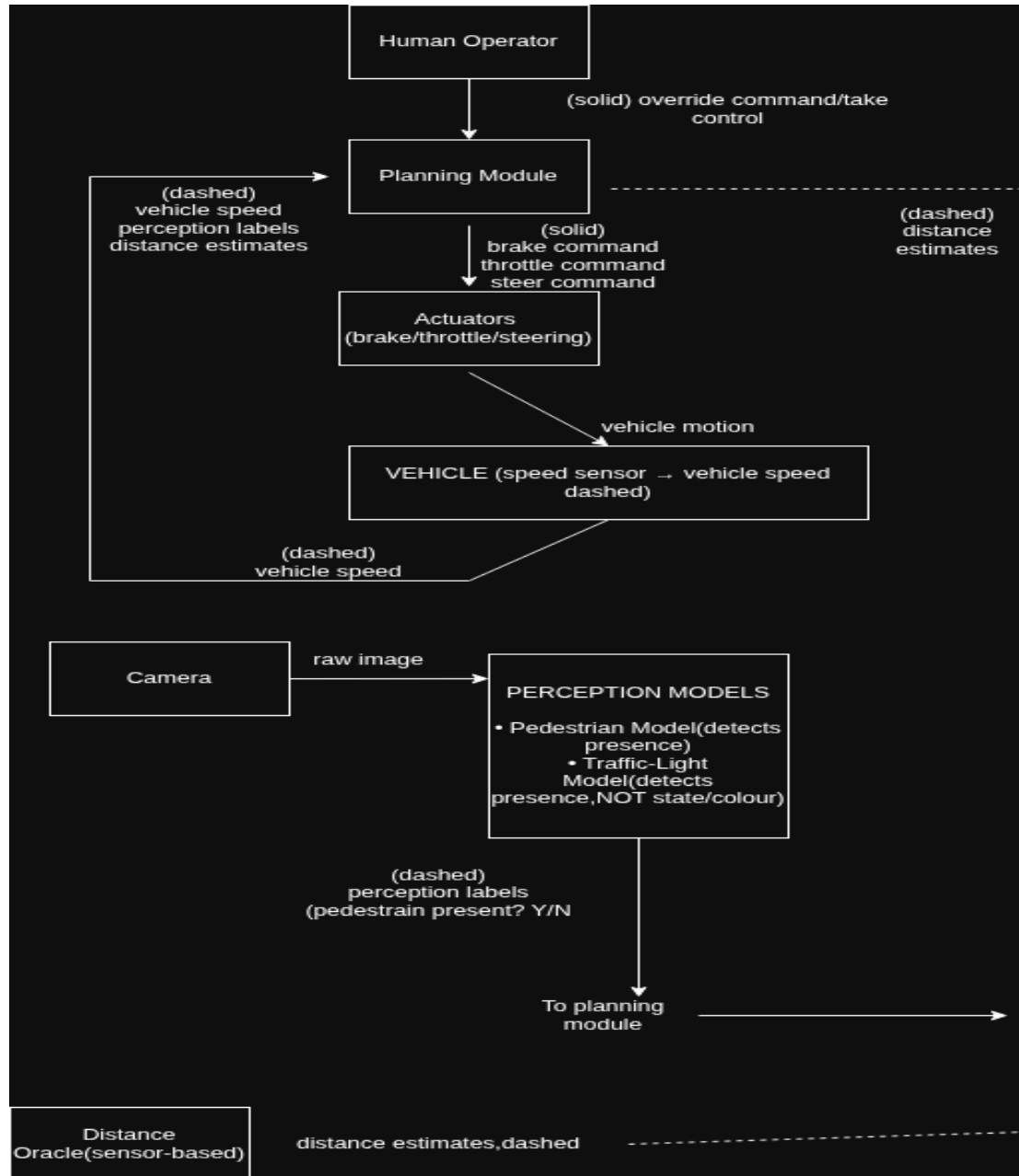
STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:

- All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).

- Control actions (solid arrows): commands flowing from controllers to controlled processes (e.g., “brake command”).
- Feedback (dashed arrows): information flowing back (e.g., “perception labels”, “distance estimates”, “vehicle speed”)



Component	Category	Role in System
Human Operator	Controller (High-Level)	Sits at the top of the hierarchy. Monitors vehicle speed and system status to decide when to issue override commands (e.g., manual control, emergency stop).
Planning Module	Controller (Low-Level)	Receives overrides from the operator and feedback from sensors. Issues direct commands to the actuators.
Perception Models	Sensor Feedback	Consumes raw camera images and sends categorical labels (pedestrians, traffic lights) back to the planner. They do not issue commands.
Distance Oracle	Sensor Feedback	Operates separately from perception models, providing direct, numerical distance estimates to the planner.
Actuators	Controlled Process	Includes the brake, throttle, and steering at the bottom of the hierarchy. They execute the planner's commands to create vehicle motion.

STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints

Exercise 2.6: Unsafe Control Actions (UCAs)

Using your control structure, identify unsafe control actions for at least two different

controllers. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., "H-1", "H-2").

For each controller, consider all four UCA types:

1. Not provided - a required action is missing.
2. Provided unsafely - the action is issued when it should not be.
3. Wrong timing / order - too early, too late, or out of sequence.
4. Wrong duration - applied too long or stopped too early.

Fill in the table below (add rows as needed):

H-1: Pedestrian is in the vehicle's path while the vehicle is moving

H-2: Another vehicle/road user is within unsafe distance while the AV is moving

H-3: Vehicle proceeds through an intersection when it should stop (traffic-light violation)

ID	Controller	Control Action	UCA Type	Hazard	Unsafe scenario
UC A 1	Planning Module	Continue driving (no brake)	Not provided	H-1	Pedestrian detector reports "pedestrian present" and distance oracle reports within stopping distance, but planner issues no brake command
UC A 2	Planning Module	Emergency brake command	Provided unsafely	H-2	Planner issues emergency brake when no pedestrian or vehicle is present (e.g., false positive from detector), causing rear-end collision from following traffic
UC A 3	Planning Module	Brake command	Wrong timing / order	H-1	Planner issues brake command too late — pedestrian has already entered critical distance zone before the command reaches actuators
UC A 4	Planning Module	Brake command	Wrong duration	H-1, H-2	Planner releases brake too early while pedestrian/vehicle is still in the path, allowing the vehicle to accelerate before the hazard has cleared

UC A 5	Human Operator	Continue driving (no brake)	Not provided	H-3	Traffic-light model reports "traffic light present" but planner has no state information (red/green), so it continues driving through a red light at intersection
UC A 6	Human Operator	Override / take manual control	Not provided	H-1, H-2	Operator fails to intervene when autonomous system makes an obviously unsafe decision (e.g., missed braking during fatigue from a 4-hour shift)
UC A 7	Human Operator	Override / take manual control	Provided unsafely	H-2	Operator takes manual control and steers or brakes incorrectly in a panic situation, causing a collision that the automated planner would have avoided
UC A 8	Human Operator	Override / take manual control	Wrong timing / order	H-1	Operator intervenes too late – grabs the wheel after the planner has already initiated emergency braking, disrupting a safe manoeuvre mid-execution

Exercise 2.7: Safety Constraints

For each UCA from Exercise 2.6, derive at least one safety constraint - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.

Classify each constraint as either:

- Model-level - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
- System-level - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

UCA	Safety Constraint	Level	Verification
UCA-1 – Planner gives no brake when pedestrian is within stopping distance	The pedestrian model must achieve recall $\geq 99\%$ on the in-distribution test set, so missed detections are rare enough that the planner reliably receives a "present" signal	Model-level	Evaluate recall on held-out test set (Sheet 4)
UCA-2 – Planner issues emergency brake with no obstacle present	The pedestrian and vehicle models must achieve precision $\geq 90\%$ to limit false alarms; additionally, the planner must require detection for at least N consecutive frames before triggering emergency	Model-level (precision) + System-level	Precision on test set; integration test of planner debounce logic

(false positive)	braking	I (frame-consistency logic)	
UCA-3 – Brake command issued too late	The full pipeline (camera → perception → planner → actuator) must complete within 100 ms end-to-end at the 99th percentile; the pedestrian model inference latency must not exceed 50 ms	Model-level (latency) + System-level (pipeline timing budget)	Stress-test latency under hardware constraints; profiling
UCA-4 – Brake released too early while hazard persists	The planner must not release braking until the pedestrian model reports "not present" for at least M consecutive frames AND distance oracle confirms safe clearance	System-level	Integration test with simulated pedestrian crossing scenarios in CARLA
UCA-5 – Vehicle continues through intersection because traffic-light state is unknown	The system must not rely solely on the traffic-light presence model for stop/go decisions at intersections; a complementary map-based or rule-based fallback (e.g., always decelerate when traffic light is detected) must be deployed	System-level	Design review; scenario testing at mapped intersections
UCA-6 – Operator fails to intervene during automated system failure	Operator monitoring protocols must include mandatory attention checks every 20 minutes; auditory alerts must be added for any perception confidence drop below threshold	System-level	Operational procedure audit; alert system integration test
UCA-7 – Operator takes control and acts unsafely	The system must not instantly transfer full authority to the operator; a 1-second blended handover window must be implemented during which the automation continues to apply limits	System-level	Human factors test; handover simulation
UCA-8 – Operator intervenes mid-manoeuve and disrupts safe braking	While emergency braking is active, operator override of the brake pedal must be suppressed or delayed by at least 500 ms unless sustained pedal input is detected	System-level	Functional safety integration test; brake override logic review

STPA Step 4: Identify Causal Loss Scenarios

Exercise 2.8: Causal Loss Scenarios

For each UCA from Exercise 2.6, identify at least one causal scenario - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. The causal factor: What component failure, misunderstanding, or adverse condition triggers the UCA?
2. Why it occurs: What is the root cause or underlying reason?
3. Connection to constraint: Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

UCA	Causal Scenario	Root Cause	Related Constraint
UCA-1 – No brake when pedestrian within stopping distance	Incorrect feedback: A pedestrian in a wheelchair or wearing unusual clothing is not in the training distribution. The pedestrian model outputs "not present." The planner receives a clear-path signal and continues at full speed.	Pedestrian model trained exclusively on sunny daytime CARLA data; out-of-distribution appearances (wheelchairs, rain gear, partial occlusion) cause false negatives	Pedestrian recall $\geq$ 99% on test set; OOD score flags > 95% of OOD samples (UCA-1 constraint)
UCA-2 – Emergency brake triggered with no real obstacle	Incorrect feedback: A sun glare or lens flare on the camera creates a bright region that the pedestrian model misclassifies as "pedestrian present." The planner fires emergency braking on an empty road, risking a rear-end collision from following traffic.	Model trained only on clean simulator images; real-world optical artefacts are out-of-distribution and cause false positives	Precision $\geq$ 90% on test set; planner requires N consecutive positive frames before triggering emergency brake (UCA-2 constraint)
UCA-3 – Brake command issued too late	Timing delay: Heavy GPU load on another inference task causes the pedestrian model to take 180 ms instead of the budgeted 50 ms. By the time the planner receives "pedestrian present" and issues the brake command, the stopping distance has already been violated.	No latency budget enforcement at model or pipeline level; single-camera system provides no early-warning redundancy	99th-percentile pipeline latency < 100 ms; pedestrian model inference $\leq$ 50 ms (UCA-3 constraint)
UCA-4 – Brake released too early while hazard persists	Flawed internal model: The pedestrian steps briefly behind a parked car (occluded), causing the detector to output "not present" for two frames. The planner's logic interprets this as hazard cleared and releases braking. The pedestrian re-enters the path and is hit.	Planner brake-release logic uses a single-frame threshold with no persistence check; occlusion is treated as absence	Planner must require M consecutive "not present" frames AND distance-oracle clearance before releasing brake (UCA-4 constraint)
UCA-5 – Vehicle continues through red light because traffic-light state is unknown	Flawed internal model: The traffic-light model correctly outputs "traffic light present" at an intersection. The planner's rule is to decelerate only if a pedestrian or vehicle is also detected nearby. With no other detections, the planner continues. The light is red.	Traffic-light model was designed only to detect presence, not colour/state; planner has no map-based or state-based fallback for intersection stop decisions	System must decelerate whenever traffic light presence is detected, regardless of other detections; map-based fallback required (UCA-5 constraint)
UCA-6 – Operator fails to intervene during automation failure	Human factor: After 3.5 hours of a 4-hour monitoring shift with no incidents, the operator's attention drifts. The pedestrian model silently fails on a rainy stretch (OOD weather), but the dashboard gives no auditory alert – only a visual confidence indicator the operator is no longer watching closely. The vehicle does not brake.	No auditory alerting; prolonged low-stimulation monitoring degrades vigilance; human is the sole fallback with no secondary check	Auditory alerts required on confidence drop below threshold; mandatory attention checks every 20 minutes (UCA-6 constraint)
UCA-7 – Operator takes	Human factor + mode confusion: The operator sees the vehicle drifting and	Abrupt authority handover with no transition window;	1-second blended handover window;

control and steers unsafely	grabs the wheel, but momentarily forgets the system is still in partial-automation mode. Their counter-steer conflicts with the planner's correction, and they over-correct into oncoming traffic.	operator not clearly informed of current automation mode at the moment of intervention	automation continues to apply steering limits during transition (UCA-7 constraint)
UCA-8 – Operator overrides brake mid-emergency manoeuvre	Human factor + timing delay: The vehicle initiates emergency braking for a pedestrian. The operator, startled, instinctively lifts their foot off what they think is a manual brake but is actually the override pedal, releasing the automated brake mid-stop at 40 km/h.	No suppression of override input during active emergency braking; operator unfamiliar with pedal function under automation	Override of brake pedal suppressed for 500 ms once emergency braking is active; sustained override input required to cancel (UCA-8 constraint)